

CS 425 : Computer Networks

Homework-3

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2. The following are server and client codes ;

```
server.py  X  client.py
server.py > ...
1  from socket import *
2
3  # server address and port
4  server_ip = "localhost"
5  server_port = 12000
6
7  # create UDP socket
8  server_socket = socket(AF_INET, SOCK_DGRAM)
9
10 # bind socket to port
11 server_socket.bind((server_ip, server_port))
12
13 print("Server waiting...")
14
15 # receive one message from client
16 message, client_address = server_socket.recvfrom(1024)
17
18 # print client message
19 print("Client:", message.decode())
20
21 # server sends one reply
22 reply = input("Server: ")
23 server_socket.sendto(reply.encode(), client_address)
24
25 # close server
26 server_socket.close()
```

The server program creates a UDP socket and binds it to the IP address **localhost** and port number **12000**. It then waits to receive one message from the client using `recvfrom()`. After receiving the message, the server prints it, takes a reply from the user, and sends the reply back to the client using `sendto()`. Finally, the server closes the socket and terminates.

```
server.py  client.py  X
client.py > ...
1  from socket import *
2
3  # server address
4  server_ip = "localhost"
5  server_port = 12000
6
7  # create UDP socket
8  client_socket = socket(AF_INET, SOCK_DGRAM)
9
10 # send message to server
11 message = input("Client: ")
12 client_socket.sendto(message.encode(), (server_ip, server_port))
13
14 # receive reply from server
15 reply, _ = client_socket.recvfrom(1024)
16
17 # print reply
18 print("Server:", reply.decode())
19
20 # close client
21 client_socket.close()
```

The client program creates a UDP socket and sends one message to the server at IP address **localhost** and port number **12000** using `sendto()`. It then waits for a reply from the server using `recvfrom()`. After receiving the reply, the client prints the message and closes the socket, ending the communication.

Instructions to Run / Testing Procedure

1. Open a terminal and navigate to the directory containing the server and client programs.

Eg: `cd udp_assignment`

2. Run the server program so that it starts listening on the specified port.

`python3 udp_server.py`

3. Open another terminal window and navigate to the same directory.

`cd udp_assignment`

4. Run the client program.

`python3 udp_client.py`

Enter a message in the client terminal when prompted.

The server receives the message and displays it. The server then sends a reply.

Observe the output on both terminals to confirm successful UDP communication.